



Matrix Inverse

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Table of contents

01

Introduction

02

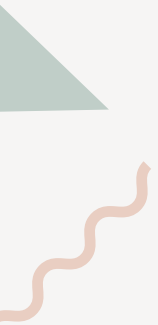
Left Inverse

03

Right Inverse

04

Square Matrix Inverse



01

Introduction to Inverse with Square Matrix



Square Matrix Inverse

Definition

For $A \in M_{n \times n}$, if there exists a matrix $B \in M_{n \times n}$ such that $AB = BA = I_n$, then:

- A is invertible (or nonsingular)
- B is the inverse of A
- The inverse of A is denoted by $B = A^{-1}$
- A square matrix that does not have an inverse is called non-invertible (or singular)

Theorem (1)

The inverse of a square matrix is unique.

- $BA=I \quad AC=I \Rightarrow BAC=(BA)C=IC=C \quad BAC=B(AC)=BI=B \Rightarrow BAC=C=B$
If B and C are both inverses of the matrix A, then $B = C$.

Square matrix inverse and column independence

Theorem (2)

A square matrix is invertible if and only if its columns are linearly independent



Proof

- Matrix is invertible $\Rightarrow$ Columns are linearly independent
- Columns are linearly independent $\Rightarrow$ Matrix is invertible

02

Left Inverse



Left Inverse

Definition

A number x that satisfies $xa = 1$ is called the inverse of a . Inverse (i.e., $\frac{1}{a}$) exists if and only if $a \neq 0$, and is unique

A matrix X that satisfies $XA = I$ is called a left inverse of A . If a left inverse exists, we say that A is left-invertible

$$A: m \times n \Rightarrow I: n \times n \Rightarrow X: n \times m$$

Example

The matrix $A = \begin{bmatrix} -3 & -4 \\ 4 & 6 \\ 1 & 1 \end{bmatrix}$

Has two different left inverses:

$$B = \frac{1}{9} \begin{bmatrix} -11 & -10 & 16 \\ 7 & 8 & -11 \end{bmatrix}$$

$$\begin{cases} BA = I \\ CA = I \end{cases}$$

$$C = \frac{1}{2} \begin{bmatrix} 0 & -1 & 6 \\ 0 & 1 & -4 \end{bmatrix}$$

Solving linear equations with a left inverse

Method

- ❑ Suppose $Ax = b$, and A has a left inverse C
- ❑ Then $Cb = C(Ax) = (CA)x = Ix = x$
- ❑ So, multiplying the right-hand side by a left inverse yields the solution

Left inverse of vector

Note

A non-zero column vector always has a left inverse.

Left inverse is not unique.

Example

$\alpha = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$ Two ways: (1) $\alpha^{-1} = \frac{1}{a_i} e_i^T$ where $a_i \neq 0$ (2) $\alpha^T \alpha = \|\alpha\|^2 \Rightarrow \frac{\alpha^T \alpha}{\|\alpha\|^2} = 1 \Rightarrow \alpha^{-1} = \frac{\alpha^T}{\|\alpha\|^2}$

Matrix with orthonormal columns $A^{-1} = A^T$

Lemma

Row vector does not have left inverse

$$\alpha = [1 \ 0 \ 3]$$

Think about $\text{rank}(BA)$, $\text{rank}(I)$ with this theory: $\text{rank}(BA) \leq \min(\text{rank}(A), \text{rank}(B))$

Left inverse and column independence

Theorem (3)

A matrix is left-invertible if and only if its columns are linearly independent

Proof

- Has left invert $\Rightarrow$ Columns are linearly independent
- Columns are linearly independent $\Rightarrow$ Has left invert

Left inverse and column independence

Note

Wide matrix is not always left invertible
Tall or square matrices can be left invertible

Example

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 3 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 & -1 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 4 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & -2 & -1 \\ 1 & 3 & 4 \\ -2 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -1 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

03

Right Inverse



Right inverse and row independence

Theorem

A matrix is right-invertible if and only if its rows are linearly independent

Proof



Right inverses

Definition

A matrix X that satisfies $AX = I$ is a right inverse of A

If a right inverse exists, we say that A is right-invertible

A is right-invertible if and only if A^T is left-invertible:

$$AX = I \Rightarrow (AX)^T = I^T \Rightarrow X^T A^T = I$$

So, we conclude:

A is right invertible if and only if its rows are linearly independent

Right-invertible matrices are wide or square

Solving linear equations with a right inverse

Method

- Suppose A has a right inverse B
- Consider the (square or underdetermined) equations of $Ax = b$
- $x = Bb$ is a solution:
- $Ax = A(Bb) = (AB)b = Ib = b$
- So $Ax = b$ has a solution for any b

Example

Same A, B, C in last example.

C^T and B^T are both right inverses of A^T

Under-determined equations $A^T x = (1, 2)$ has (different) solutions.

$$B^T(1, 2) = \left(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right), \quad C^T(1, 2) = \left(0, \frac{1}{2}, -1\right)$$

there are many other solutions as well

Conclusion: Left and Right Inverse



Linear equations and matrix inverse

Definition

Left-Invertible matrix: if X is a left inverse of A , then

$$Ax = b \Rightarrow x = XAx = Xb$$

There is at most one solution using X (if there is a solution, it must be equal to Xb)

We must know in advance that there exists at least one solution

Why “at most”??

$$XA = I$$

$$\begin{cases} -y_1 + y_2 = -4 \\ 0y_1 - y_2 = 3 \\ 2y_1 + y_2 = 0 \end{cases}$$

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -1 \\ 2 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 5 & 2 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -1 & 1 & -4 \\ 0 & -1 & 3 \\ 2 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{array} \right]$$

Linear equations and matrix inverse

Note

□ If the system of equations $Ax = b$ is consistent, and if a matrix B exists such that $BA = I$, then the system of equations has a unique solution, namely $x = Bb$.

□ **Right-inversible matrix:** if X is a right inverse of A , then there is at least one solution ($x=Xb$):

$$x = Xb \Rightarrow Ax = AXb = b$$

□ To pursue these ideas further, suppose that again we want to solve a system of linear equations, $Ax = b$. Assume now that we have another matrix, B , such that $AB = I$. Then we can write $A(Bb) = (AB)b = Ib = b$; hence Bb solves the equations $Ax = b$. This conclusion did not require a prior assumption that a solution exist; we have produced a solution. The argument does not reveal whether Bb is the only solution. There may be others.

□ **Invertible matrix:** if A is invertible, then

$$Ax = b \Leftrightarrow x = A^{-1}b$$

There is a unique solution

Conclusion

- System of linear equations $Ax = b$:
 - A right inverse of A , say $AB = I$. Then Bb is a solution, as is verified by nothing $A(Bb) = (AB)b = Ib = b$.
 - Why don't need to check the consistency for using right inverse?
 - A left inverse of A , say $CA = I$, then we can only conclude that Cb is the sole candidate for a solution; however, it must be checked by substitution to determine whether, in fact, it is a solution

04

Square Matrix Inverse



Inverse

Definition

For $A \in M_{n \times n}$, if there exists a matrix $B \in M_{n \times n}$ such that $AB = BA = I_n$, then:

- A is invertible (or nonsingular)
- B is the inverse of A
- The inverse of A is denoted by $B = A^{-1}$
- A square matrix that does not have an inverse is called non-invertible (or singular)
- **For a square matrix left and right inverse are the same. Rows and columns are linear independent.**

Square matrix inverse

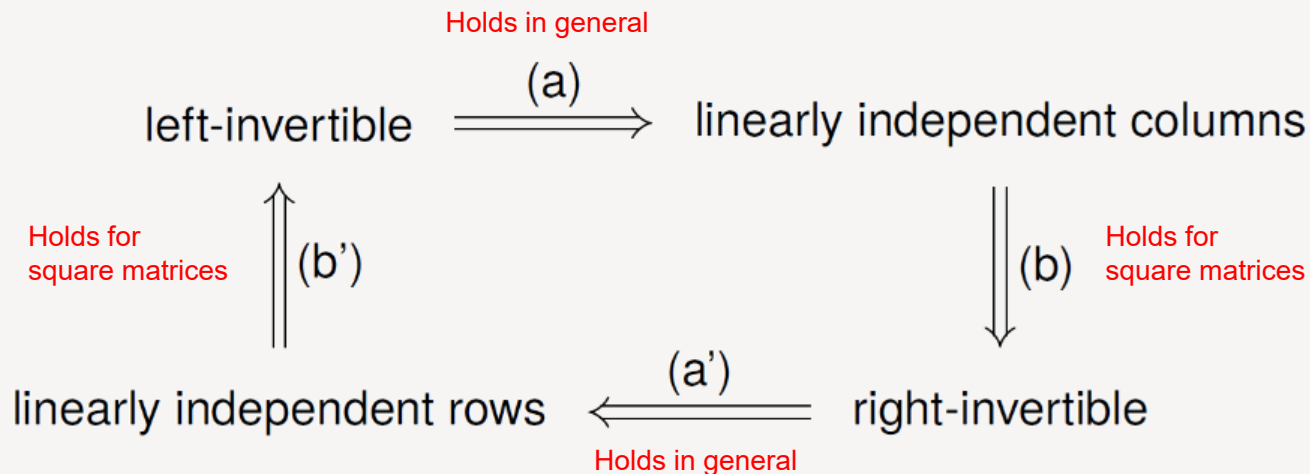
Theorem

For a square matrix, the right and left inverse are the same

Proof



Invertible (**Nonsingular**) Matrices



Gauss-Jordan Elimination for finding the Inverse of a matrix

Method

- ❑ Let A be a $n \times n$ matrix:
 - ❑ Adjoin the identity $n \times n$ matrix I_n to A to form the matrix $[A : I_n]$.
 - ❑ Compute the reduced echelon form of $[A : I_n]$.
- ❑ If the reduced echelon form is of the type $[I_n : B]$, then B is the inverse of A .
- ❑ If the reduced echelon form is not the type $[I_n : B]$, in that the first $n \times n$ submatrix is not I_n then A has no inverse.

$$[A \mid I] \text{ Gauss-Jordan elimination } [I \mid A^{-1}]$$

Important

An $n \times n$ matrix is invertible if and only if its reduced echelon form is I_n .

A is row equivalent to I_n

Inverse (Example)

Example

Find inverse of the following matrix using Gauss-Jordan Elimination:

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$$

$$AX = I \Rightarrow \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x_{11} + 4x_{21} & x_{12} + 4x_{22} \\ -x_{11} - 3x_{21} & -x_{12} - 3x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

By equating corresponding entries we have:

$$\begin{cases} x_{11} + 4x_{21} = 1 \\ -x_{11} - 3x_{21} = 0 \end{cases} \quad (1)$$
$$\begin{cases} x_{12} + 4x_{22} = 0 \\ -x_{12} - 3x_{22} = 1 \end{cases} \quad (2)$$

This two system of linear equations have the same coefficient matrix, which is exactly the matrix A

Inverse (Example)

Rest of The Example

Using Gauss-Jordan Elimination on the matrix A with the same row operations

$$\begin{aligned} (1) &\Rightarrow \left[\begin{array}{cc|c} 1 & 4 & 1 \\ -1 & -3 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 1 & 1 \end{array} \right] \Rightarrow x_{11} = -3, x_{21} = 1 \\ (2) &\Rightarrow \left[\begin{array}{cc|c} 1 & 4 & 0 \\ -1 & -3 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & 0 & -4 \\ 0 & 1 & 1 \end{array} \right] \Rightarrow x_{12} = -4, x_{22} = 1 \end{aligned}$$

Thus $X = A^{-1} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}$

$$\left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ -1 & -3 & 0 & 1 \end{array} \right] \xrightarrow{\text{Gauss-Jordan elimination}} \left[\begin{array}{cc|cc} 1 & 0 & -3 & -4 \\ 0 & 1 & 1 & 1 \end{array} \right]$$

$A \qquad I \qquad I \qquad A^{-1}$

Solution for $\begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix}$

Solution for $\begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix}$

Elementary Matrices

Definition

Each Elementary Matrix E is invertible. The inverse of E is the elementary matrix of the same type that transforms E back into I .

Example

Find the inverse of $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$

Inverse

Definition

Properties (If A is invertible matrix, k is a positive integer and c is a scalar):

A^{-1} is invertible and $(A^{-1})^{-1} = A$

A^k is invertible and $(A^k)^{-1} = A^{-k} = (A^{-1})^k$

cA is invertible if $c \neq 0$ and $(cA)^{-1} = \frac{1}{c}A^{-1}$

A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$

Theorem

If A and B are invertible matrices of order n , then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(A_1A_2A_3 \cdots A_n)^{-1} = A_n^{-1} \cdots A_3^{-1}A_2^{-1}A_1^{-1}$$

Inverse

Theorem

The solution set K of any system $Ax=b$ of m linear in n unknowns is (so is a linear map T with standard matrix A), s is a particular solution:

$$K = s + \text{Null}(T_A)$$



Theorem (Using above Theorem)

Let $Ax = b$ be a system of n linear equations in n variable.

The system has exactly one solution Cb if and only if A is invertible and $C = A^{-1}$.



Invertible Matrix

Definition

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible

Note

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. $\det A = ad - bc$.

2×2 matrix A is invertible if and only if $\det A \neq 0$.

Invertible (Nonsingular) matrices

Conclusion

The following are equivalent for a square matrix A :

- ☐ A is invertible
- ☐ Columns of A are linearly independent
- ☐ Rows of A are linearly independent
- ☐ A has a left inverse
- ☐ A has a right inverse

$$\text{row rank}(A) = \text{col rank}(A) = n$$

If any of these hold, all others do

Properties

Properties

- $(AB)^{-1} = B^{-1}A^{-1}$
- If A is nonsingular, then A^T is nonsingular
 $(A^T)^{-1} = (A^{-1})^T$ (sometimes denoted A^{-T})
- **Negative matrix powers:** $(A^{-1})^k$ is denoted by A^{-k}
- With $A^0 = I$, Identity $A^k A^l = A^{k+l}$ holds for any integers k, l

Triangular matrices

Theorem

Lower Triangular L with non-zero diagonal entries is invertible

Proof??

Theorem

Upper Triangular R with non-zero diagonal entries is invertible

Proof??



Why Matrix of Change of Basis is invertible?

Because the column and rows of it is the basis so they are linear independent and invertible



Rank and Inverse

Theorem

Given a square matrix M and its inverse M^{-1} , then M and M^{-1} have the same rank.



Rank and Inverse

Theorem

If A is $m \times n$ and B is an $n \times n$ invertible matrix, then $\text{rank}(AB) = \text{rank}(A)$.

Solution:

By theorem $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B)) \Rightarrow$

$$\text{rank}(AB) \leq \text{rank}(A) \quad \text{rank}(AB) \leq \text{rank}(B)$$

Also, we can write:

$$AB B^{-1} = A \Rightarrow \text{rank}(AB B^{-1}) \leq \text{rank}(AB) \Rightarrow \text{rank}(A) \leq \text{rank}(AB)$$

so, using $[\ast]$, $[\ast\ast]$ then $\text{rank}(A) = \text{rank}(AB)$

QUESTION 10. (5 points) Suppose that $A \in M_{m,n}(\mathbb{C})$

(a) Give the definition of a *middle inverse* of A .

Solution: A middle inverse of A is a matrix $B \in M_{n,m}(\mathbb{C})$ such that

$$ABA = A \quad \text{and} \quad BAB = B.$$

(b) Suppose that A is right-invertible. Show that B is a middle inverse of A if and only if B is a right inverse of A .

Solution: Suppose B is a right inverse of A . Then

$$ABA = IA = A \quad \text{and} \quad BAB = BI = B.$$

So B is a middle inverse of A . Now suppose that B is a middle inverse of A . Since A is right invertible, it has some right inverse C . Then

$$ABA = A \implies ABAC = AC \implies AB = I \implies AB = I.$$

Hence B is a right inverse of A .

(c) Give the definition of the *pseudoinverse* of A .

Solution: The pseudoinverse of A is the unique $A^+ \in M_{n,m}(\mathbb{C})$ such that A^+ is a middle inverse of A and both AA^+ and A^+A are hermitian.

(d) Suppose $A = P\Sigma Q^H$ is a singular value decomposition for A , with

$$\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}_{m \times n}, \quad D = \text{diag}(d_1, d_2, \dots, d_r), \quad d_1, d_2, \dots, d_r \in \mathbb{R}_{>0}.$$

Give an explicit expression for A^+ .

Solution: We have $A^+ = Q\Sigma'P^H$, where

$$\Sigma^{-1} = \begin{bmatrix} D^{-1} & 0 \\ 0 & 0 \end{bmatrix}_{n \times m}.$$

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following are equivalent.

- A is an invertible matrix.
- A is nonsingular matrix.
- A is row equivalent to the $n \times n$ identity matrix.
- A has n pivot positions.
- The equation $Ax = 0$ has only the trivial solution.
- The columns of A form a linearly independent set.
- The linear transformation $x \rightarrow Ax$ is one-to-one.
- The equation $Ax = b$ has at least one solution for each $b \in \mathbb{R}^n$.
- The columns of A span $\mathbb{R}^n$.
- The linear transformation $x \rightarrow Ax$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- There is an $n \times n$ matrix C such that $CA = I$.
- There is an $n \times n$ matrix D such that $AD = I$.
- A^T is an invertible matrix.

Resources

- https://math.berkeley.edu/~arash/54/notes/02_03.pdf
- <https://www.ijsr.net/archive/v4i9/SUB156717.pdf>
- <https://math.colorado.edu/~nita/SystemsofLinearEquations.pdf>

